

Hussein Naser

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WORK EXPERIENCE

Carleton University - Canada, Ottawa, ON

University Teaching/Research Assistant

[4 Jan 2023 - Current]

- Conduct research in nonlinear control systems, autonomous unmanned aerial vehicles (UAVs), cooperative aerial transportation, and human-UAV physical interaction.
- Develop mathematical models, simulation environments, and control algorithms for UAV and aerial robotic systems using MATLAB, ROS2, Gazebo, and related engineering tools.
- Design and implement simulation and experimental platforms for validating control, state-estimation, and human-robot interaction methodologies.
- Develop force/torque sensing and estimation frameworks for physical human-UAV interaction, reducing reliance on dedicated force-torque sensors.
- Implement state estimation, sensor fusion, and filtering techniques, including Kalman filter-based approaches and nonlinear observers, for improved system performance and robustness.
- Analyze experimental and simulation data to evaluate controller performance, estimation accuracy, and system stability under external disturbances.
- Support undergraduate engineering education by teaching laboratory sessions, demonstrating engineering concepts, and mentoring students in project-based learning activities.
- Assist with course delivery, laboratory development, student assessment, and technical guidance for undergraduate engineering courses.

Romaeris Corporation - Canada, Ottawa

Research engineer (Internship)

[1 May 2025 - 30 Aug 2025]

- Developed a 6-DOF mathematical dynamic model of a powered unmanned hang glider aircraft, enabling simulation-based controller design and flight performance evaluation.
- Designed and implemented a model-based flight control system to improve vehicle stability, trajectory tracking, and autonomous operation capabilities.
- Built a comprehensive MATLAB/Simulink simulation environment for evaluating aircraft dynamics, control performance, and mission scenarios prior to hardware testing.
- Collaborated with multidisciplinary engineering teams to support the development of next-generation unmanned aerial vehicle (UAV) technologies.
- Generated technical documentation and engineering reports to communicate modeling assumptions, control design methodology, and simulation results.

Quality Assurance Department/ University of Thi-Qar - Iraq, Nasiriyah

Manager of Training and Technical Support Unit

[16 Jan 2019 - 22 Dec 2022]

- Led training programs and technical support services for university staff and faculty.
- Coordinated the annual performance evaluation process for faculty members and employees.
- Managed institutional data collection and reporting for national and international university rankings.
- Served as the primary liaison with international ranking organizations and external stakeholders.

Biomedical Engineering Department, University of Thi-Qar - Iraq, Nasiriyah

University lecturer

[15 Sep 2018 - 22 Dec 2022]

- Delivered undergraduate engineering courses in control systems, electronics, and related engineering disciplines.
- Supervised senior undergraduate capstone projects and provided academic mentoring.
- Participated in many committees as a faculty member.

UNITAR Hiroshima - Japan, Hiroshima

Entrepreneurship coach (Online)

[2 Jan 2020 - 29 Jan 2021]

- Facilitated online international training sessions and structured discussion workshops in entrepreneurship and professional development.
- Assessed trainee performance, assignments, and learning progress for Iraqi and international participants.
- Provided coaching, guidance, and individualized support to help trainees achieve learning objectives and project milestones.
- Coordinated virtual learning activities and ensured engagement in remote training environments.

Mechatronics Engineering Department, University of Baghdad - Iraq, Baghdad

University lecturer

[1 Sep 2015 - 14 Sep 2018]

- Delivered undergraduate engineering courses, supporting curriculum instruction and student learning outcomes.
- Supervised and advised senior undergraduate capstone design projects in engineering disciplines.
- Participated in many committees as a faculty member.

AL-Khwarizmi College of Engineering / University of Baghdad - Iraq, Baghdad

Mechatronics engineer

[18 Jan 2009 - 14 Aug 2012]

- Supported multiple engineering laboratories including robotics, electronics circuits, microprocessors, sensors and instrumentation, and computer systems.
- Assisted in designing and supervising laboratory experiments for undergraduate engineering students.
- Provided technical guidance to students during laboratory sessions and hands-on implementation of experiments.
- Maintained, calibrated, and troubleshooted laboratory instruments and experimental equipment to ensure operational reliability.
- Participated in departmental committees for planning and procurement of laboratory equipment for Al-Khwarizmi College of Engineering.

EDUCATION & TRAINING

PhD Candidate in Mechanical and Aerospace Engineering

Carleton University [4 Jan 2023 - Current]

City: Ottawa | Country: Canada

Master of Sciences in Mechanical Engineering

College of Engineering, Michigan State University [27 Aug 2013 - 19 Aug 2015]

City: Lansing | Country: United States

Certificate in English as Second Language

ESL, Michigan State University [29 Aug 2012 - 9 Aug 2013]

City: Lansing | Country: United States

Bachelor of Sciences in Mechatronics Engineering

AL-Khwarizmi College of Engineering / University of Baghdad [15 Nov 2003 - 15 Jul 2008]

City: Baghdad | Country: Iraq

Machine Learning

Coursera [10 Jun 2020 - 20 Sep 2020]

Website: coursera.org

UNITAR Hiroshima Empowering Social Entrepreneurs and Youth Leaders for Iraq

UNITAR Hiroshima [20 Sep 2019 - 14 Feb 2020]

City: Hiroshima | Country: Japan

Certificate in Tech Entrepreneurship

Five One Labs [7 Jul 2019 - 1 Aug 2019]

City: Sulaimani | Country: Iraq

Android Developer Track

Udacity, 1 Million Arab Coders Initiative [1 Nov 2018 - 30 Nov 2018]

City: Online | Country: United Arab Emirates

Data Analyst Track

Udacity, 1 Million Arab Coders Initiative [1 Aug 2018 - 31 Aug 2018]

City: Online | Country: United Arab Emirates

Educational Qualification Course

Development and Continuing Education Center, University of Baghdad [4 Apr 2016 - 21 Apr 2016]

City: Baghdad | Country: Iraq

Computers & Printers Maintenance

Computer Center, University of Baghdad [31 Oct 2010 - 23 Nov 2010]

City: Baghdad | Country: Iraq

Automation Robotics Training System 5150 & Pneumatic I Training System 6081

Pyramid of Knowledge Co, Lab-Volt Company Agent [1 Oct 2010 - 18 Oct 2010]

City: Baghdad | Country: Iraq

PUBLICATIONS

Publications

• Journal Papers:

1. Hussein N. Naser, Hashim A Hashim, Mojtaba Ahmadi, Quaternion-based unscented Kalman filter for robust wrench estimation of human-UAV physical interaction, Signal Processing, 2026.
2. Hussein N. Naser, Hashim A Hashim, Mojtaba Ahmadi, Aerial assistive payload transportation using quadrotor UAVs with nonsingular fast terminal SMC for human physical interaction, Results in Engineering, 2025.
3. Hussein N. Naser, and H. A. Hashim, sEMG-based hand gestures classification using a semi-supervised multi-layer neural networks with Autoencoder, Systems and Soft Computing, 2024.
4. Asaad Migot, Hussein NH Naser, Adnan A Ugla, Victor Giurgiutiu, Investigations of fatigue crack detection using local vibration techniques: Numerical and experimental studies, Journal of Nondestructive

Evaluation, 2023.

5. Hussein Naeem Hasan Naser, "Kreen" an Electronic Platform for Towing Service as an Alternative Solution for Writing on Walls Phenomenon in Iraq, International Journal of Computing and Digital Systems, 2022.
6. Hussein Naeem Hasan, A Cost Effective Deaf-Mute Electronic Assistant System Using Myo Armband and Smartphone, International Journal of Science and Research (IJSR), 2017.

• **Conference Papers:**

1. Hussein N. Naser, Hashim A Hashim, Mojtaba Ahmadi, Adaptive gain nonlinear observer for external wrench estimation in human-uav physical interaction, In Proc. of the 2026 IEEE International Conference on Robotics & Automation (ICRA), Vienna, Austria 2026.
2. Hussein N. Naser, Hashim A Hashim, Mojtaba Ahmadi, Human Physical Interaction based on UAV Cooperative Payload Transportation System using Adaptive Backstepping and FNTSMC, 2025 IEEE American Control Conference (ACC), July 8-10, 2025, Denver, CO, USA.
3. Hussein N. Naser, Husam Hammood, Asaad Q Migot, Internet-Based Smartphone System for After-Stroke Hand Rehabilitation, 2023 International Conference on Engineering, Science and Advanced Technology (ICESAT), Mosul, Iraq.
4. Hussein Naeem Hasan, A Wearable Rehabilitation System to Assist Partially Hand Paralyzed Patients in Repetitive Exercises*, Journal of Physics: Conference Series, Volume 1279, First International Scientific Conference Al-Ayen University 30–31 March 2019, Thi-Qar, Iraq.

Research Interests

1. Adaptive and robust nonlinear control.
2. Assistive aerial cooperative payload transportation.
3. Human-UAVs physical interaction.
4. Forces and torques estimation for human-robot interaction.
5. Human guidance, navigation, and control.
6. Gesture Recognition, and Wearable Hand Rehabilitation.

NETWORKS AND MEMBERSHIPS

- Member of Iraqi Engineers Union (2009 – until now).
- IEEE Student member (2026).

CONFERENCES & SEMINARS

Conferences Participation and Attending :

1. IEEE International Conference on Robotics & Automation (ICRA), Vienna, Austria 2026.
2. American Control Conference (ACC), July 8-10, 2025, Denver, Colorado, USA.
3. International Conference on Engineering, Science and Advanced Technology (ICESAT), June 21, 2023, Mosul, Iraq.
4. The 1st International Conference on Advances in Engineering Science and Technology (AEST-2021), April 26-27, 2021, Babylon, Iraq.
5. 1st International Scientific Conference of Al-Ayen University, March 2019, Nasiriyah, Thi-Qar, Iraq.
6. The First International Conference on Modern Trend in Applied Sciences, November 19-21, 2017, Langkawi Island, Malaysia.
7. 7th and the 1st international scientific engineering conference. May 2017, University of Baghdad, Iraq.
8. Postgraduate studies conference. December 2017, College of Engineering, University of Baghdad, Baghdad, Iraq.

Seminars Participation :

1. 3D printing technology and its applications/ AL-Khwarizmi College of Engineering (2018).
2. The symposium on the launch of Baghdad smart city project. Mayoralty of Baghdad (2018).
3. Myo gestures control armband and its applications/ AL-Khwarizmi College of Engineering/ Mechatronics Dept. (2018).
4. Robotic Fish as a mechatronic application/ AL-Khwarizmi College of Engineering (2017).

HONOURS AND AWARDS

1. Scholarship to complete PhD degree from the University of Thi-Qar/ Ministry of Higher Education and Scientific Research, Iraq (2023).
2. Innovation shield award for graduation project supervision from Mechatronics Engineering Department/ University of Baghdad (2018).
3. Certificates of recognition, president of University of Baghdad (2018) and president of University of Thi-Qar (2019, 2020, 2021).
4. Scholarship to complete master degree HCED office of the Iraqi prime minister (2012).
5. Graduate Project Competition Prize, Ministry of Higher Education and Scientific Research, Iraq (2009).
6. Iraqi Excelled Student Reward (Waedoun) Iraqi Vice President, Iraq (2008).

PROJECTS

[6 Jan 2020 - 30 Apr 2020]

Fully Automatic Sterilization Tunnel Designed and implemented a fully automatic sterilization tunnelfor the University of Thi-Qar and Thi-Qar Health Directorate to support efforts in combating COVID-19.

[13 Jan 2014 - 31 Dec 2015]

Autonomous Robotic Fish-human Interaction System Designed and implemented an Autonomous Robotic Fish-Human Interaction Systemfor the Michigan State University Museum exhibit.

TEACHING

Carleton University (Teaching Assistant)

Mechanical and Aerospace Engineering Department, Faculty of Engineering & Design (Ottawa, Canada).

1. Engineering Dynamics, (MAAE2101), Fall 2025, Fall 2024, Fall 2023, Winter 2023.
2. Visual Communication, (ECOR1047), Winter 2025, Winter 2024.
3. Dynamics, (ECOR1048), Winter 2026, Winter 2025, Winter 2024.

University of Thi-Qar (Instructor)

Biomedical Engineering Department, College of Engineering (Nasiriyah, Thi-Qar, Iraq).

1. Analog Electronics, 2021.
2. Control Systems Design, 2020.

University of Baghdad (Instructor)

Mechatronics Engineering Department, AL-Khwarizmi College of Engineering (Baghdad, Iraq).

1. Artificial Neural Networks fundamentals, 2017, 2018, 2019.
2. Fuzzy Logic fundamentals, 2017, 2018, 2019.
3. Engineering Drawing Software (SolidWorks), 2017.
4. AC Electric Machines fundamentals, 2016.
5. DC Electric Machines fundamentals, 2016.

PROJECTS SUPERVISING

Undergraduate Projects

1. **Supervised** undergraduate project: " Follow me robotic platform," College of Engineering, University of Thi-Qar (2019-2020).
2. **Supervised** undergraduate project: "Conversion of gestures and movements of deaf-mute people to audible speech using myo armband and smartphone," AL-Khwarizmi College of Engineering , University of Baghdad, (2017-2018) .
3. **Supervised** undergraduate project: "Design and implementation of a 3D printed mobile Robot," AL-Khwarizmi College of Engineering , University of Baghdad, (2016-2017) .

4. **Supervised** undergraduate project: "Design an android Application to control a mobile Robot,"
AL-Khwarizmi College of Engineering , University of Baghdad, (2016-2017) .

PROFESSIONAL ACTIVITIES

Journal Reviewer

- IEEE Transactions on Consumer Electronics
- IEEE Journal of Biomedical and Health Informatics
- Journal of Intelligent & Robotic Systems
- Intelligent Service Robotics
- Journal of Circuits, Systems, and Computers (JCSC)
- International Journal of Computing and Digital Systems (IJCDS)
- Al-Nahrain Journal for Engineering Sciences

MEDIA COVERAGE

Unitar news stories (Developing Entrepreneurship Skills in Iraq, Hussein Naeem Hasan Naser), 2020

Link: <https://unitar.org/about/news-stories/stories/hussein-naeem-hasan-naser-0>

TECHNICAL SKILLS

Technical Skills

1. **Robotics & Autonomy:** ROS2, Gazebo Simulation, PX4 Autopilot, ArduPilot, MAVROS, Mission Planner, QGroundControl, Pixhawk Systems.
2. **Control & Estimation:** PID Control, Kalman Filtering, Sensor Fusion, State Estimation, Nonlinear Observers, Nonlinear Control Systems.
3. **Programming & Software Development:** Python, C/C++, MATLAB, Simulink, PHP, MySQL, Android Development (Android Studio).
4. **Embedded Systems & Hardware:** Embedded Systems Design, Microcontrollers (Arduino), Single-Board Computers (Raspberry Pi), Sensor Integration.
5. **Aerospace & UAV Systems:** Flight Control Systems, UAV Dynamics, Autonomous Systems, Drone Simulation and Testing.
6. **Communication Systems:** Wireless Communication Modules (RC systems, XBee, Bluetooth).
7. **Mechanical & Design Tools:** SolidWorks, Autodesk Fusion 360, CAD Modeling, 3D Printing.
8. **Electronics & PCB Design:** PCB and Circuit Design using Eagle.
9. **Development & Productivity Tools:** Linux, Microsoft Office Suite, LaTeX.

LANGUAGE SKILLS

Mother tongue(s): Arabic

Other language(s): English (fluent)